

LocalShed

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Project Summary

This document outlines the full product specification, user personas, technical requirements, and financial estimates for LocalShed.

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EXECUTIVE SUMMARY:

{{EXECUTIVE_SUMMARY}}

USER PERSONAS:

Persona 1:

Priya Sharma

Age: 35

Background: Priya is a homemaker in Gurgaon who designs and stitches custom blouses and creates intricate embroidery work. She sells to friends and family but wants to reach more customers in her apartment complex and nearby areas to supplement her family income.

Tech Comfort: She is comfortable using WhatsApp and basic social media apps but finds complex e-commerce interfaces intimidating.

Primary Goal: To earn an additional ₹15,000-₹20,000 per month by selling her handmade products without compromising her family responsibilities.

Key Frustrations: Limited customer reach beyond her immediate social circle. High commissions on existing online platforms make them unaffordable. Difficulty in managing orders and payments efficiently.

Key Needs: An easy-to-use platform to list products with clear photos. Direct messaging with customers. Simple payment and order management. Assurance of local customer base.

Quote: "I make beautiful things, but how do I show them to people in my building? WhatsApp is too messy."

Persona 2:

Anjali Verma

Age: 29

Background: Anjali is a marketing professional living in Noida who appreciates authentic, handcrafted items.

She actively seeks unique home decor and jewelry that tell a story and wants to support local artisans.

Tech Comfort: She is highly proficient with mobile apps and online shopping platforms.

Primary Goal: To discover and purchase unique, high-quality handmade products from local sellers in her neighborhood or residential complex.

Key Frustrations: Difficulty in finding genuine local artisans amidst mass-produced goods. Lack of transparency in product origin and seller authenticity.

Inconvenience of traveling to distant markets or dealing with complex shipping.

Key Needs: A curated feed of local products. Ability to message sellers directly for customization.

Trustworthy platform with seller reviews. Easy discovery of items within a close radius.

Quote: "I love finding unique treasures made by people nearby. It feels more special and supports our community."

Persona 3:

Amit Kumar

Age: 40

Background: Amit is a small-scale administrator for a residential society in Gurgaon. He is responsible for managing community services and ensuring residents have access to local amenities and trusted services.

Tech Comfort: Comfortable with basic administrative software and mobile communication tools.

Primary Goal: To provide residents with a safe and convenient platform to discover and transact with local micro-entrepreneurs, fostering a sense of community and local economy.

Key Frustrations: Residents often complain about the lack of access to local services or unique products.

Managing informal vendor activities within the society can be challenging. Ensuring quality and trust for resident-to-resident transactions.

Key Needs: A simple way to onboard and manage local sellers. A trusted platform that vets sellers and products. A system for community announcements or featured sellers. Easy dispute resolution mechanisms.

Quote: "If we can help our residents buy and sell locally, it strengthens our society. We need a reliable way to do this."

USER STORIES:

[PRIMARY USER TYPE STORIES]

- As a saree seller, I want to upload high-quality photos of my sarees with detailed descriptions, so that customers can see the product accurately.
- As an artisan, I want to set my own prices for my pottery, so that I can maximize my profit.
- As a handmade jewelry maker, I want to set my delivery radius, so that I only receive orders from nearby customers.
- As a seller, I want to receive direct messages from potential buyers, so that I can answer their questions and build trust.
- As a home decor maker, I want to mark my products as 'ready for pickup', so that customers know they can collect them quickly.
- As a seller, I want to view my sales history and earnings, so that I can track my business performance.
- As a seller, I want to receive push notifications for new messages and orders, so that I don't miss any opportunities.
- As a seller, I want to have a simple profile page with my story and product categories, so that customers can learn more about me.

- As a seller, I want to be able to easily update product availability, so that I don't sell out-of-stock items.
- As a seller, I want to offer discounts on my products, so that I can attract more customers.

[SECONDARY USER TYPE STORIES]

- As a customer, I want to browse products from sellers within a 2km radius, so that I can find local artisans easily.
- As a customer, I want to filter products by category (e.g., sarees, jewelry, home decor), so that I can find what I'm looking for quickly.
- As a customer, I want to message a seller directly to ask about customization or availability, so that I can get personalized service.
- As a customer, I want to see seller ratings and reviews, so that I can make informed purchasing decisions.
- As a customer, I want to view seller profiles to understand their craft and story, so that I can connect with them.
- As a customer, I want to see products listed within my residential complex, so that I can arrange for easy pickup.
- As a customer, I want to see estimated delivery or pickup times, so that I know when to expect my order.

- As a customer, I want to be able to save products or sellers to my favorites list, so that I can revisit them later.
- As a customer, I want to receive notifications when a seller I follow lists new products, so that I don't miss out.
- As a customer, I want to pay securely via UPI, so that I have a convenient payment option.

[ADMIN/MODERATOR STORIES]

- As an admin, I want to view a list of all registered sellers and customers, so that I can manage the user base.
- As an admin, I want to be able to moderate reported content or users, so that I can maintain platform integrity.
- As an admin, I want to view transaction data and commission reports, so that I can track revenue.
- As an admin, I want to be able to send push notifications to all users or specific segments, so that I can make announcements.
- As an admin, I want to manage seller subscription plans and track renewals, so that I can manage monetization.
- As an admin, I want to access a dashboard to monitor key platform metrics (e.g., active users,

transactions, revenue), so that I can assess performance.

FEATURE LIST:

MVP (Must Build for Launch)

- **User Registration & Profile Creation:** Allows sellers to create profiles with basic details and customers to register, enabling personalized experiences.
- **Product Listing & Management:** Sellers can upload product images, descriptions, prices, and categories to showcase their items.
- **Hyper-local Search & Discovery:** Customers can find sellers and products within a specified radius (e.g., 2km) or by residential complex.
- **Direct In-App Messaging:** Enables real-time communication between buyers and sellers for inquiries and negotiation.
- **Transaction Processing (UPI via Razorpay):** Securely facilitates payments for products through integrated UPI options.
- **Order Management (Seller View):** Sellers can view incoming orders, update status (e.g., accepted, ready for pickup/delivery), and track fulfillment.
- **Order Management (Customer View):** Customers can view their placed orders, track status, and see pickup/delivery details.

- **Push Notifications:** Alerts users about new messages, orders, and important updates to ensure timely engagement.
- **Basic Seller Dashboard:** Provides sellers with an overview of their listings, orders, and earnings.

Phase 2 (After Key Milestone)

- **Seller Ratings & Reviews:** Allows customers to rate sellers and products, building trust and accountability.
- **Wishlist/Favorites:** Enables customers to save products or sellers they are interested in for future reference.
- **Featured Listings for Sellers:** A basic mechanism for sellers to pay for increased visibility.
- **Location-based Delivery Options:** Simple options for sellers to indicate if they offer local delivery within a certain radius.
- **Admin Moderation Tools:** Enhanced tools for admins to manage disputes, block users, and ensure platform safety.
- **Basic Analytics Dashboard for Sellers:** Provides insights into views, inquiries, and sales trends.

Future (Long-term Roadmap)

- Multi-language Support: Expand to include more regional Indian languages beyond Hindi and English.
- Integrated Shipping Options: For sellers looking to expand beyond hyper-local, offer curated shipping partnerships.
- Community Forums/Groups: Create spaces for sellers to share tips and for buyers to discover new trends.
- Promotional Tools for Sellers: Allow sellers to create coupons or run limited-time offers.
- AI-powered Product Recommendations: Suggest products to customers based on their browsing history and preferences.

WIREFRAME DESCRIPTIONS:

Screen 2: Onboarding/Welcome Screen

Purpose: Introduce the app and its value proposition, guiding new users through initial setup.

Key Elements: App logo, compelling tagline, carousel of benefits (e.g., 'Shop Local', 'Support Artisans'), 'Sign Up' button, 'Login' button.

User Action: Tap 'Sign Up' or 'Login', or swipe through carousel.

Screen 3: Sign Up/Login Screen

Purpose: Allow users to create an account or log in to an existing one.

Key Elements: Phone number input field, 'Get OTP' button, OTP input field, 'Verify OTP' button, options for Google/Facebook login (optional), 'Already have an account? Login' link.

User Action: Enter phone number, receive and enter OTP, or select social login.

Screen 4: Seller/Customer Role Selection

Purpose: Determine the user's primary role on the platform.

Key Elements: Two prominent buttons: 'I want to Sell Products' and 'I want to Buy Products'. Brief descriptions under each button.

User Action: Tap on the relevant role button.

Screen 5: Seller Profile Setup

Purpose: Enable sellers to create their public profile.

Key Elements: Profile picture upload, Seller Name input, Business Name input (optional), 'About Me/My Story' text area, Location (auto-detected or manual input), Category selection (e.g., Sarees, Jewelry, Pottery).

User Action: Upload photo, fill in details, select categories.

Screen 6: Product Listing Form

Purpose: Allow sellers to add new products to their catalog.

Key Elements: Product image upload (multiple), Product Name input, Description text area, Price input, Category selection, Stock Quantity input, 'Add Product' button.

User Action: Upload images, fill in product details, set price and quantity.

Screen 7: Homepage (Customer View)

Purpose: Display a feed of products from local sellers.

Key Elements: Search bar at the top, location indicator (e.g., 'Near You in Sector 15, Gurgaon'), filter/sort options, scrollable list of product cards (image, name, price, seller name, distance).

User Action: Scroll through products, tap on a product card, use search or filters.

Screen 8: Product Detail Page

Purpose: Show comprehensive information about a specific product.

Key Elements: Large product image carousel, Product Name, Price, Detailed Description, Seller Name (link to profile), Location/Distance, 'Message Seller' button, 'Add to Cart' (for future) or 'Buy Now' (for MVP).

User Action: View details, message seller, initiate purchase.

Screen 9: Chat Screen

Purpose: Facilitate direct communication between buyer and seller.

Key Elements: Header with seller/buyer name, message input field, 'Send' button, chat history display area (messages with timestamps).

User Action: Type and send messages.

Screen 10: Seller Dashboard

Purpose: Provide sellers with an overview of their sales and orders.

Key Elements: Total Earnings, Number of Orders, List of active orders with status, 'Add New Product' button, Link to edit profile.

User Action: View sales summary, manage orders, add new products.

Screen 11: Payment Confirmation/Order Summary

Purpose: Confirm the successful transaction and display order details.

Key Elements: Success message ('Payment Successful!'), Order ID, Product details, Seller information, Total amount paid, Delivery/Pickup instructions, 'View Order Details' button.

User Action: Review order details, proceed to order history.

DATA MODELS:

Table: Users

Fields: user_id (UUID) - unique identifier |
phone_number (String) - verified mobile number |
name (String) - display name | email (String, optional) -
user's email | profile_picture_url (String, optional) -
URL to profile image | role (Enum: 'SELLER',
'CUSTOMER', 'ADMIN') - user's primary role |
created_at (Timestamp) - account creation time |
updated_at (Timestamp) - last profile update time
Relationships: One-to-many with Products (if seller),
One-to-many with Orders (if customer)

Table: Sellers

Fields: seller_id (UUID) - unique identifier | user_id
(UUID) - foreign key to Users table | business_name
(String, optional) - seller's brand name | description
(Text) - seller's story/bio | location (GeoPoint) -
geographical coordinates of seller's base |
selling_radius (Integer) - max distance for local sales
in km | commission_rate (Decimal) - commission
percentage for this seller (can override global) |
is_premium (Boolean) - flag for premium subscription |
subscription_expiry_date (Timestamp, optional) - end

date of premium subscription | created_at (Timestamp)
| updated_at (Timestamp)

Relationships: One-to-one with Users, One-to-many
with Products, One-to-many with Orders

Table: Products

Fields: product_id (UUID) - unique identifier | seller_id
(UUID) - foreign key to Sellers table | name (String) -
product name | description (Text) - detailed product
description | price (Decimal) - selling price | currency
(String) - e.g., 'INR' | images (Array of Strings) - URLs
to product images | category (String) - e.g., 'Sarees',
'Jewelry' | stock_quantity (Integer) - number of items
available | availability (Boolean) - true if product is
available | created_at (Timestamp) | updated_at
(Timestamp)

Relationships: Many-to-one with Sellers, One-to-many
with OrderItems

Table: Orders

Fields: order_id (UUID) - unique identifier |
customer_id (UUID) - foreign key to Users table |
seller_id (UUID) - foreign key to Sellers table |
total_amount (Decimal) - total price of the order |
status (Enum: 'PENDING', 'ACCEPTED', 'REJECTED',
'READY_FOR_PICKUP', 'DELIVERED', 'CANCELLED') -

current_order_status | payment_status (Enum: 'PENDING', 'PAID', 'FAILED') - status of payment | payment_transaction_id (String, optional) - Razorpay transaction ID | order_date (Timestamp) - date and time order was placed | pickup_delivery_details (Text, optional) - notes on pickup/delivery arrangement | created_at (Timestamp) | updated_at (Timestamp)
Relationships: Many-to-one with Users (customer), Many-to-one with Sellers, One-to-many with OrderItems

Table: OrderItems

Fields: order_item_id (UUID) - unique identifier | order_id (UUID) - foreign key to Orders table | product_id (UUID) - foreign key to Products table | quantity (Integer) - number of this product in the order | price_at_order (Decimal) - price of the product when ordered | created_at (Timestamp)

Relationships: Many-to-one with Orders, Many-to-one with Products

Table: Chats

Fields: chat_id (UUID) - unique identifier | order_id (UUID, optional) - associated order ID if chat is related to an order | seller_id (UUID) - foreign key to Sellers

table | customer_id (UUID) - foreign key to Users table
| created_at (Timestamp) | updated_at (Timestamp)
Relationships: One-to-many with Messages,
Many-to-one with Orders (optional)

Table: Messages

Fields: message_id (UUID) - unique identifier | chat_id (UUID) - foreign key to Chats table | sender_id (UUID) - foreign key to Users table | content (Text) - message text | timestamp (Timestamp) - time message was sent | read_status (Boolean) - flag indicating if message has been read

Relationships: Many-to-one with Chats

Table: SellerSubscriptions

Fields: subscription_id (UUID) - unique identifier | seller_id (UUID) - foreign key to Sellers table | plan_name (String) - e.g., 'Premium' | start_date (Timestamp) | end_date (Timestamp) | status (Enum: 'ACTIVE', 'EXPIRED', 'CANCELLED') | created_at (Timestamp) | updated_at (Timestamp)

Relationships: Many-to-one with Sellers

API ENDPOINT LIST:

AUTH ENDPOINTS:

- POST /api/v1/auth/register — Register a new user (phone number, OTP verification)
- POST /api/v1/auth/login — Login an existing user (phone number, OTP verification)
- POST /api/v1/auth/verify-otp — Verify OTP for registration/login
- POST /api/v1/auth/resend-otp — Resend OTP for a given phone number

PRIMARY USER (SELLER) ENDPOINTS:

- POST /api/v1/sellers/profile — Create or update seller profile
- GET /api/v1/sellers/profile — Get seller profile
- POST /api/v1/products — Create a new product listing
- GET /api/v1/products — Get list of seller's products
- GET /api/v1/products/{product_id} — Get details of a specific product
- PUT /api/v1/products/{product_id} — Update a product listing
- DELETE /api/v1/products/{product_id} — Delete a product listing

- GET /api/v1/sellers/dashboard — Get seller dashboard summary (orders, earnings)
- GET /api/v1/sellers/orders — Get list of orders for a seller
- PUT /api/v1/sellers/orders/{order_id}/status — Update order status (e.g., ACCEPTED, READY_FOR_PICKUP)

SECONDARY USER (CUSTOMER) ENDPOINTS:

- GET /api/v1/products — Search and browse products (with filters for location, category, etc.)
- GET /api/v1/products/{product_id} — Get details of a specific product
- POST /api/v1/chats — Initiate a chat with a seller
- GET /api/v1/chats — Get list of chats for a customer
- GET /api/v1/chats/{chat_id} — Get messages within a specific chat
- POST /api/v1/chats/{chat_id}/messages — Send a message in a chat
- GET /api/v1/customers/orders — Get list of orders placed by a customer
- POST /api/v1/orders — Create a new order

ADMIN ENDPOINTS:

- GET /api/v1/admin/users — Get list of all users
- GET /api/v1/admin/sellers — Get list of all sellers

- GET /api/v1/admin/products — Get list of all products
- GET /api/v1/admin/orders — Get list of all orders
- PUT /api/v1/admin/users/{user_id}/status — Activate/Deactivate a user
- POST /api/v1/admin/notifications — Send broadcast notification

PAYMENT ENDPOINTS:

- POST /api/v1/payments/create-order — Create a payment order with Razorpay
- POST /api/v1/payments/webhook — Handle Razorpay payment webhook for confirmation

TECH STACK

RECOMMENDATION:

Frontend / Mobile:

Component	Technology	Reasoning
Mobile App	React Native	Enables cross-platform development for both iOS and Android from a single codebase, significantly reducing development time and cost, which is crucial for an MVP with a limited budget.
Web Frontend	{{STACK_WEB_FRONTEND}}	{{STACK_WEB_FRONTEND_REASONING}}
Stack Management		

Backend & Infrastructure:

Component	Technology	Reasoning
Backend API		
Database		
Caching		
Hosting		

Third-Party Services:

Service	Technology	Reasoning
Backend API		
Database		
Caching		
Hosting		

COST & TIMELINE ESTIMATE:

Development Cost Breakdown:

Role	Hours	Rate (₹/hr)	Total (₹)
Backend Developer (Node.js)	600	1200	720,000
Mobile Developer (React Native)	700	1300	910,000
UI/UX Designer	200	1000	200,000
Project Manager/Tech Lead	150	1500	225,000
QA Tester	150	800	120,000
Total Development Cost: ₹2,175,000			

First Year Additional Costs:

Hosting & Infrastructure (Year 1):	₹120,000
Third-party APIs & Services (Razorpay, Twilio, Google Maps):	₹80,000
Legal & Compliance Setup (DPDP, T&Cs):	₹50,000
QA & Testing Tools:	₹30,000

Total Additional Costs: ₹280,000

Total First Year Budget: ₹2,455,000

Timeline: 6 months with 1 Backend Developer, 1 Mobile Developer (React Native), 1 UI/UX Designer, 1 QA Tester, Part-time Project Manager/Tech Lead developers

HIRING RECOMMENDATION:

Recommended Approach:

Given the budget of ₹2-5 lakh for MVP development and a timeline of 3-6 months, hiring a full-time in-house team is not feasible. The most practical approach is to leverage a combination of skilled freelancers and a small, agile development agency. Freelancers can be sourced from platforms like Upwork, Fiverr, or Toptal for specialized roles like UI/UX design and specific backend tasks. For the core mobile and backend development, consider engaging with a reputable Indian tech agency that has experience in building marketplace apps. This allows for faster team formation and project execution. Focus on hiring individuals or agencies with proven experience in building scalable mobile applications, preferably with a track record in the Indian startup ecosystem and familiarity with the recommended tech stack. Prioritize candidates who demonstrate strong communication skills and an understanding of the hyper-local market dynamics.

Where to Find Developers:

- Upwork: Skilled freelance backend (Node.js) and mobile (React Native) developers. Expect rates from ₹1000-₹2500/hr depending on experience.
- Fiverr: UI/UX designers and QA testers. Rates can vary widely, from ₹500-₹1500/hr.
- Toptal: Top-tier developers, but can be more expensive. Suitable for critical roles if budget allows.
- LinkedIn: For identifying and connecting with agencies or senior freelance developers in India.
- Indian Tech Communities (e.g., local developer meetups, specific forums): For finding niche talent or smaller agencies.

Red Flags to Watch:

- Lack of specific examples for hyper-local or marketplace apps: If a developer can only show generic CRUD apps or social media clones, they might not grasp the nuances of location-based services and direct seller-buyer interactions.
- Unwillingness to discuss technical challenges or trade-offs: A good developer will articulate potential issues and propose solutions, not just agree to everything.

- Poor communication or slow response times during the interview process: This is a strong indicator of how they will perform during development, especially with time-sensitive tasks.
- Inability to explain their development process or testing methodologies: Suggests a lack of structured approach, increasing the risk of bugs and delays.
- Over-reliance on generic solutions without understanding the specific problem: For instance, suggesting a complex, global shipping integration when the core is hyper-local pickup.
- Vague or inflated time/cost estimates without clear breakdowns: Indicates a lack of project planning or an attempt to overpromise.

RISKS & COMPLIANCE:

Key Risks:

Sr. No	Risk Description	Mitigation Strategy
1.		
2.		
3.		
4.		